Benchmark against Suncal

Generated by `PYTHONHASHSEED=0 python3 tools/research/benchmarks/suncal_compare.py` (research log D74, extended in D93). Every library project that Suncal can express is evaluated by both programs: measurement models with one output (including correlated inputs and intermediate quantities), models with several outputs (JCGM 102), calibrations with uncertainties in both variables (ISO 6143) and conformity assessments (JCGM 106). The projects that are not compared, and why, are listed at the end; the other project types (top-down evaluation, proficiency testing, precision experiments, control charts, sampling, consensus values, error functions of weighing instruments) have no counterpart in Suncal.

Monte Carlo standard deviations (the program's trials and seed; Suncal 10^6 draws, NumPy seed 1) are compared by their relative difference and by that difference divided by its sampling standard error $\sqrt{((\kappa + 2)/(4N_1) + (\kappa + 2)/(4N_2))}$, κ the excess kurtosis of the output (N_1 the program's trials, $N_2 = 10^6$ Suncal's; the kurtosis term corrects the standard error of a standard deviation for non-normal outputs, and is taken as 0 for models with several outputs). The end of the comparison gives the probability that the largest of that many independent standard normal |z| values is at least as large as the one observed, so the dispersion can be judged against sampling error alone (review 2026-11-08, item 15).

Suncal 1.7.1; 171 projects compared (135 measurement models, 12 multivariate models, 6 calibrations, 18 conformity assessments).

1. Measurement models

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
bipm-isrd-01-qnmr-dmso2-ma-d2o	—	0.99726	-1.5e-13	0.0007321	-1.1e-16	—	—
bipm-isrd-01-qnmr-dmtp-ma-dmso-d6	—	0.999426	+0.0e+00	0.0004096	-2.2e-16	—	—
bipm-isrd-02-qnmr-dmtp-dmso2-cdcl3	—	0.999063	+0.0e+00	0.0005038	+0.0e+00	—	—
bipm-isrd-02-qnmr-ma-dmso2-d2o	—	0.99976	+0.0e+00	0.000788	-1.1e-16	—	—
bipm-isrd-06-qnmr-ba-btmsb-d4-cd3od	—	0.999774	-3.0e-13	0.0007397	-3.3e-16	+1.68e-03	+1.7
bipm-isrd-06-qnmr-dmso2-btmsb-d4-dmso-d6	—	0.996722	+1.4e-13	0.0007951	+2.2e-16	—	—
bipm-isrd-07-qnmr-dmso2-dss-d6-d2o	—	0.996246	+9.4e-14	0.001186	+0.0e+00	—	—
bipm-isrd-07-qnmr-ma-dss-d6-d2o	—	0.998687	+1.0e-13	0.001085	+2.2e-16	—	—
bipm-isrd-08-qnmr-btmsb-d4-ba-cd3od	—	0.998818	+3.6e-13	0.0003119	+2.2e-16	—	—
emue-e3.4-soil-lead-iso21748-rms-bias	—	27.8	-4.7e-14	1.434	-1.9e-15	-1.34e-03	-1.3
emue-e3.4-soil-lead-iso21748	—	27.8	-4.8e-14	1.397	-1.9e-15	-9.93e-04	-1.0
emue-e3.8-ammonia-permeation	intermediates	70.7959	+0.0e+00	0.5978	+0.0e+00	-7.32e-04	-0.7

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
gum5-e1-ph-temperature-interpolation-correlated	intermediates, correlations	7.00435	+0.0e+00	0.004083	+0.0e+00	-2.38e-03	-2.4
gum5-e1-ph-temperature-interpolation	intermediates	7.00435	+0.0e+00	0.003222	+0.0e+00	+2.54e-05	+0.0
gum5-e1-ph-two-point-25c-correlated-standards	correlations	7.00023	+0.0e+00	0.005132	+0.0e+00	+1.31e-03	+1.3
gum5-e1-ph-two-point-25c	—	7.00023	+0.0e+00	0.004083	+0.0e+00	+6.46e-04	+0.6
gum5-e2-benzo-a-pyrene	correlations	0.0140283	+0.0e+00	0.002019	-2.2e-16	—	—
gum5-e3-molar-mass-glucose	—	180.156	+0.0e+00	0.00381	+2.2e-16	-5.05e-04	-0.7
gum5-e5-ghg-inventory-agriculture-shared-activity	—	4282.14	+0.0e+00	835.2	+0.0e+00	-2.15e-04	-0.2
gum5-e5-ghg-inventory-agriculture	—	4282.14	+0.0e+00	835.2	-1.1e-16	-3.93e-04	-0.4
iso13528-e5-los-angeles-reference-value	—	23.3475	+0.0e+00	0.3534	+0.0e+00	—	—
iso21748-c2-meat-content	—	95.637	+0.0e+00	1.99	+0.0e+00	+6.29e-04	+0.6
iupac2024-4a-water-hardness	—	96.6462	+0.0e+00	3.177	+0.0e+00	-1.06e-03	-1.1
iupac2024-4h-cadmium-standard-beta-mcm	—	1.0027	+0.0e+00	0.0008637	-1.3e-14	—	—
iupac2024-4i-argentometric-titration	—	10	+0.0e+00	0.02554	+0.0e+00	+5.62e-04	+0.6
iupac2024-5c-benzoic-acid-purity-beta	—	0.99992	+0.0e+00	3e-05	+0.0e+00	—	—
nist-tn1900-e8-molecular-weight-co2	—	44.0094	+0.0e+00	0.0007182	+2.2e-16	-6.62e-04	-0.8
nist-tn1900-e9-cadmium-standard	—	1002.7	+0.0e+00	0.8637	+1.1e-15	+5.09e-04	+0.5
possolo-meija-2022-plasma-anion-gap	—	25	+0.0e+00	1.848	+0.0e+00	-3.38e-04	-0.3
quam-a1-cadmium-standard-volume	intermediates	1002.7	+0.0e+00	0.8352	-4.4e-16	-4.09e-04	-0.5
quam-a1-cadmium-standard	—	1002.7	+0.0e+00	0.8637	-5.6e-16	+4.39e-04	+0.4
quam-a2-naoh-standardisation-submodels	intermediates	0.102136	+1.2e-12	0.0001007	-1.5e-12	-1.20e-03	-1.2
quam-a2-naoh-standardisation	—	0.102136	+0.0e+00	9.864e-05	-1.1e-16	+3.57e-04	+0.4
quam-a3-hcl-titration	—	0.101387	+0.0e+00	0.0001828	+0.0e+00	-3.95e-04	-0.4
quam-a4-pesticides-combination	—	1.11111	+0.0e+00	0.3771	+0.0e+00	+4.58e-04	+0.4
quam-a5-cadmium-release	—	0.0150646	+0.0e+00	0.00147	+0.0e+00	+2.67e-04	+0.3
quam-a7-assay-standard	—	0.0926048	+5.0e-13	2.777e-05	+2.2e-16	-2.15e-04	-0.2
quam-a7-lead-double-idms	intermediates	0.0537375	+7.7e-14	0.00018	-1.1e-16	+1.11e-03	+1.1
ufs-a6-cadmium-soil	—	0.3192	+1.9e-15	0.02943	+2.2e-16	—	—
ufs-a6-phosphorus-soil	—	115.8	+0.0e+00	12.99	-1.1e-16	—	—

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
ume-procedure-19-2-qnmr-purity-internal-standard	—	995	-7.3e-14	1.567	+4.4e-16	—	—
ea402-s3-resistor-10kohm	—	1.00002e+07	+0.0e+00	8.328	-6.4e-12	—	—
ea402-s6-power-sensor-18ghz	—	0.933044	-6.9e-15	0.01618	-6.7e-16	—	—
ea402-s7-step-attenuator-30db	—	30.0433	+0.0e+00	0.02222	+0.0e+00	—	—
ea402-s9-dmm-100v	—	0.1	+0.0e+00	0.02957	+0.0e+00	-3.96e-04	-0.6
gum-h2-impedance	correlations	254.26	+0.0e+00	0.2363	+2.2e-16	-1.02e-03	-1.0
gum-h2-reactance	correlations	219.847	+0.0e+00	0.2956	+0.0e+00	-5.26e-04	-0.5
gum-h2-resistance	correlations	127.732	+2.0e-13	0.07107	+8.9e-16	+1.17e-03	+1.2
jcgm101-9.4-comparison-loss-x000-correlated-higher-order	correlations	50	+0.0e+00	0	+0.0e+00	-2.06e-03	-0.9
jcgm101-9.4-comparison-loss-x000-correlated	correlations	0	+0.0e+00	0	+0.0e+00	+8.55e-04	+0.4
jcgm101-9.4-comparison-loss-x000	—	50	+0.0e+00	0	+0.0e+00	+1.22e-03	+0.7
jcgm101-9.4-comparison-loss-x010-correlated-higher-order	correlations	150	+5.0e-01	100	+0.0e+00	-9.38e-04	-0.5
jcgm101-9.4-comparison-loss-x010-correlated	correlations	100	+0.0e+00	100	+0.0e+00	-2.01e-03	-1.2
jcgm101-9.4-comparison-loss-x010	—	150	+5.0e-01	100	+0.0e+00	+2.50e-04	+0.2
jcgm101-9.4-comparison-loss-x050-correlated-higher-order	correlations	2550	+1.0e-01	500	+0.0e+00	+6.27e-04	+0.7
jcgm101-9.4-comparison-loss-x050-correlated	correlations	2500	+0.0e+00	500	+0.0e+00	-2.75e-04	-0.3
jcgm101-9.4-comparison-loss-x050	—	2550	+1.0e-01	500	-1.1e-16	-8.93e-04	-1.0
m3003-c5-dvm-dominant-rectangular	—	1	+0.0e+00	0.3053	+0.0e+00	-1.85e-03	-1.5
m3003-k1-resistor-10kohm	—	10000.1	+0.0e+00	0.004454	+0.0e+00	—	—
m3003-k2-power-sensor-18ghz	—	93.155	+0.0e+00	1.693	+0.0e+00	—	—
m3003-k3-attenuator-30db-mismatch-e5	—	30.05	+0.0e+00	0.02022	+0.0e+00	—	—
m3003-k3-attenuator-30db	—	30.05	+0.0e+00	0.02457	+0.0e+00	—	—
m3003-l3-dmm-1v-range-095v	—	0	+0.0e+00	5.367	-1.1e-16	—	—
m3003-l3-dmm-1v-range-1v-dominant	—	0	+0.0e+00	5.629	+0.0e+00	—	—
m3003-l3-dmm-1v-range-500mv	—	0	+0.0e+00	3.087	+0.0e+00	—	—
num-10-resistance	correlations	127.732	+2.0e-13	0.06998	-1.0e-15	+1.90e-03	+1.9
possolo-meija-2022-wheatstone-bridge	—	956.993	+0.0e+00	6.958	+0.0e+00	+2.36e-04	+0.2

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
ume-procedure-19-1-dmm-10v-calibration	—	212	+0.0e+00	35.09	+2.2e-16	—	—
cg19-ex1-flask-1000ml	intermediates	999.622	+6.8e-12	0.0168	+2.2e-16	—	—
cg19-ex2-pycnometer-100ml	intermediates	100.014	+0.0e+00	0.0009002	+0.0e+00	—	—
ea402-s12-water-meter-error	—	0.000335126	+0.0e+00	0.0006813	+2.2e-16	-3.72e-04	-0.4
ea402-s12-water-meter-mean-error	—	0.001	+0.0e+00	0.0009097	+0.0e+00	—	—
ea402-s12-water-meter-volume	—	199.933	+0.0e+00	0.1089	+0.0e+00	-1.52e-04	-0.2
m3003-k9-flowrate-standard-conditions	intermediates	12.5255	+6.5e-15	0.2746	+2.2e-16	—	—
nist-tn1900-e4-pitot-tube	—	57.4804	+0.0e+00	0.348	+2.2e-16	+2.14e-03	+2.1
gum5-e9-torque-slope-ols	—	1.01065	+0.0e+00	0.001521	-1.1e-16	—	—
gum5-e9-torque-slope-wls	—	1.00849	+0.0e+00	0.0007652	+0.0e+00	—	—
jcgm101-9.2.2-additive-gaussian	—	0	+0.0e+00	2	+0.0e+00	+4.60e-04	+0.5
jcgm101-9.2.3-additive-rectangular	—	0	+0.0e+00	2	+0.0e+00	-4.84e-05	-0.1
jcgm101-9.2.4-additive-dominant-rectangular	—	0	+0.0e+00	10.15	+2.2e-16	+7.29e-04	+1.3
num-3-ratio-abc	—	28.8	+0.0e+00	8.689	+2.2e-16	-2.36e-03	-1.6
ea402-s10-vernier-calliper	—	100	+0.0e+00	32.34	-3.1e-15	-2.23e-04	-0.3
ea402-s13-ring-gauge-90mm	—	0.232296	-1.4e-16	0.404	+0.0e+00	—	—
ea402-s13-ring-gauge-temperature	intermediates	0	+0.0e+00	0.1467	+0.0e+00	-7.92e-04	-0.9
ea402-s4-gauge-block-50mm	—	4.99999e+07	+0.0e+00	32.18	+0.0e+00	+7.99e-05	+0.1
gum-h1-end-gauge	intermediates	5.00008e+07	+0.0e+00	31.67	+0.0e+00	+1.20e-03	+1.2
jcgm101-9.5-gauge-block-approximate	—	838	+0.0e+00	32.14	+0.0e+00	—	—
jcgm101-9.5-gauge-block-nonlinear	—	838	+0.0e+00	32.14	+0.0e+00	—	—
m3003-k10-sheet-thickness	—	1.514	+0.0e+00	0.009632	+0.0e+00	—	—
m3003-k6-gauge-block-10mm-temperature-model	—	9.99994	+0.0e+00	4.102e-05	-1.1e-16	—	—
m3003-k6-gauge-block-10mm	—	9.99994	+0.0e+00	4.074e-05	+0.0e+00	—	—
num-8-end-gauge-gaussian	—	5.00008e+07	+0.0e+00	31.71	+0.0e+00	+5.25e-04	+0.6
possolo-meija-2022-storage-tank-volume	—	7204.3	+0.0e+00	53.75	+0.0e+00	+4.13e-04	+0.4
cg18-h1-100g-option2	—	0.000697877	+3.4e-11	0.0001485	+2.2e-16	—	—
cg18-h1-220g-max-option1-delta-t	—	0.0013	+0.0e+00	0.0004912	+0.0e+00	—	—
cg18-h1-220g-max-option1	—	0.0013	+0.0e+00	0.001971	+0.0e+00	—	—

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
cg18-h1-220g-max-option2	—	0.00129533	+1.2e-11	0.0002262	+2.2e-16	—	—
cg18-h1-220g-zero	—	0	+0.0e+00	0.0001176	+2.2e-16	—	—
cg18-h2-60kg-10kg-option1	—	0	+0.0e+00	1.552	+0.0e+00	—	—
cg18-h2-60kg-max-adjusted	—	0	+0.0e+00	5.725	+0.0e+00	—	—
cg18-h2-60kg-max-option1-delta-t	—	-10	+0.0e+00	5.944	+0.0e+00	—	—
cg18-h2-60kg-max-option1	—	-10	+0.0e+00	5.973	+0.0e+00	—	—
cg18-h2-60kg-max-option2	—	-10.0013	+0.0e+00	5.938	+0.0e+00	—	—
cg18-h2-60kg-zero	—	0	+0.0e+00	1.238	+0.0e+00	—	—
cg18-h3-30t-10t-weights	—	10	+0.0e+00	7.98	+0.0e+00	—	—
cg18-h3-30t-15t-substitution	—	15	+0.0e+00	14.6	+0.0e+00	—	—
cg18-h3-30t-max-substitution	—	30	+0.0e+00	23.86	-3.3e-16	—	—
cg18-h3-30t-zero	—	0	+0.0e+00	6.749	+0.0e+00	—	—
ea402-s2-weight-10kg	—	25	+0.0e+00	29.26	+0.0e+00	+1.75e-04	+0.2
ilac-p14-cmc-floor-weight-10kg	—	25	+0.0e+00	29.26	+0.0e+00	-1.97e-04	-0.2
jcgm101-9.3-mass-calibration-higher-order	—	1.234	+0.0e+00	0.05385	+0.0e+00	+5.68e-04	+0.6
jcgm101-9.3-mass-calibration	—	1.234	+0.0e+00	0.05385	+0.0e+00	-7.62e-04	-0.9
m3003-k4-weight-10kg	—	10000	+0.0e+00	0.02456	+2.2e-16	—	—
m3003-k5-weighing-machine-200g	—	-0.1	+0.0e+00	0.1528	+0.0e+00	—	—
nist-tn1900-e1-weighing	—	50.284	+0.0e+00	0.007113	+0.0e+00	-1.17e-03	-1.2
gum5-e14-radioactive-decay-bi207	—	10.81	+0.0e+00	0.05411	+0.0e+00	-3.92e-04	-0.4
gum5-e15-wire-rope-breaking-force	—	10005.7	+0.0e+00	9.912	+2.2e-16	—	—
nist-tn1900-e13-thermal-expansion-gaussian-95	—	17.2662	-9.1e-14	1.755	+0.0e+00	+4.06e-05	+0.1
nist-tn1900-e13-thermal-expansion-gaussian-99	—	17.2662	-9.1e-14	1.755	+0.0e+00	-4.78e-04	-0.6
nist-tn1900-e3-falling-ball-viscometer	—	5.68741	-8.6e-16	1.027	+0.0e+00	-7.43e-04	-0.6
num-11-stefan-boltzmann	—	5.67037	-2.6e-10	2.046e-05	-6.7e-16	+5.26e-05	+0.1
emue-e2.2-vacuum-gauge-correction-applied	—	1.00245	+0.0e+00	0.003551	+0.0e+00	—	—
emue-e2.2-vacuum-gauge-correction-ignored	—	1.00195	+0.0e+00	0.002965	+0.0e+00	—	—
emue-e2.2-vacuum-gauge-correction-in-uncertainty	—	1.00195	+0.0e+00	0.003563	-1.1e-16	—	—
emue-e2.2-vacuum-gauge-sum-model-correlated	—	-0.6282	+8.2e-15	0.9087	+0.0e+00	—	—

Project	features	estimate	Δy/u	u (first order)	u/u_Suncal − 1	MCM s.d./Suncal MC − 1	z
emue-e2.2-vacuum-gauge-sum-model-correlation-ignored	—	-0.6282	+6.5e-15	1.14	+0.0e+00	—	—
m3003-k8-pressure-indicator-2mpa	—	16.9932	-1.8e-12	43.01	+2.2e-16	—	—
ea402-s11-block-calibrator-180c	—	180100	+0.0e+00	164.3	+0.0e+00	-7.49e-04	-0.9
ea402-s5-thermocouple-emf	—	36228.8	+0.0e+00	24.97	+6.7e-16	-3.72e-05	-0.0
ea402-s5-thermocouple-furnace	—	1000.5	+0.0e+00	0.6409	+0.0e+00	+3.07e-04	+0.4
m3003-k11-gas-temperature	—	22.9681	+0.0e+00	0.08634	+1.6e-15	—	—
m3003-k11-probe-calibration	—	0.432	-6.6e-15	0.04225	+5.6e-15	—	—
m3003-k7-thermocouple-emf	intermediates	36228.8	+0.0e+00	25.84	+0.0e+00	—	—
m3003-k7-thermocouple-furnace	—	1000.5	+0.0e+00	0.6409	+0.0e+00	—	—
nist-tn1900-e2-surface-temperature	—	25.5909	+0.0e+00	0.8725	+2.2e-16	—	—

max |u/u_Suncal − 1| = 6.4e-12; max |Δy|/u = 5.0e-01; with correlations or intermediates: 24; MC compared: 68, max |Δs.d.|/s.d. = 2.4e-03, max |z| = 2.4

2. Models with several outputs (JCGM 102)

Project	outputs	max Δy/u	max	u/u_Suncal − 1		max
gum5-e4-gravimetric-gas-mixture	8	4.9e-11	4.4e-16	2.2e-16	—	—
iupac2024-4b-silicon-isotopic-abundances	3	0.0e+00	2.2e-16	2.2e-16	-2.21e-03	-1.3
possolo-meija-2022-atomic-weight-lead-hipb1	5	6.8e-10	1.6e-14	3.1e-14	+2.94e-03	+1.7
jcgm102-9.3-polar-x001-correlated	2	0.0e+00	0.0e+00	1.1e-16	+4.80e-04	+0.5
jcgm102-9.3-polar-x001	2	0.0e+00	2.2e-16	0.0e+00	+1.38e-03	+1.4
jcgm102-9.3-polar-x010-correlated	2	0.0e+00	0.0e+00	2.2e-16	+5.49e-04	+0.5
jcgm102-9.3-polar-x010	2	0.0e+00	0.0e+00	1.1e-16	+1.48e-03	+1.5
jcgm102-9.3-polar-x100-correlated	2	0.0e+00	0.0e+00	2.2e-16	+1.01e-03	+1.0
jcgm102-9.3-polar-x100	2	0.0e+00	0.0e+00	2.2e-16	+1.37e-03	+1.4
jcgm102-9.2.2-additive-gaussian	2	0.0e+00	0.0e+00	2.2e-16	-1.32e-03	-1.3
jcgm102-9.2.3-additive-rectangular	2	0.0e+00	0.0e+00	2.2e-16	+9.89e-04	+1.0
jcgm102-9.2.4-additive-dominant-rectangular	2	0.0e+00	0.0e+00	2.2e-16	-2.94e-04	-0.3

max |u/u_Suncal − 1| = 1.6e-14; max |Δr| = 3.1e-14

3. Calibration with uncertainties in both variables (ISO 6143)

Project	function	Suncal method	Γ	max	Δb	/u(b)	max	u(b)/u_Suncal(b) − 1
emue-e3.3-no2-analyser-dynamic-dilution-covariances-ignored	linear	York	1.27	5.0e-09	7.5e-03	8.1e-09	1	1.1e-03
iso6143-b1-no-uv-linear	linear	York	0.57	3.7e-11	2.5e-04	3.2e-09	3	1.0e-04
iso6143-b2-n2-natural-gas-linear	linear	York	1.63	2.2e-13	1.1e-04	1.4e-10	2	7.3e-05
iso6143-b2-n2-natural-gas-quadratic	quadratic	ODR	0.87	1.8e-04	1.1e-04	6.3e-05	2	1.9e-04
iso6143-b3-no-ndir-exponential	exponential	ODR	0.35	5.3e-06	4.8e-05	1.4e-06	1	4.6e-05
iso6143-b3-no-ndir-power	power	ODR	1.16	6.5e-06	7.3e-03	1.6e-06	1	7.0e-04

Γ is the largest weighted deviation of a calibration point (ISO 6143 goodness of fit). Suncal's York and ODR covariances are the Gauss–Newton approximation; KAPSAM propagates the input uncertainties through the minimum with the full Hessian (ISO 6143 A.2, A.6, A.7), which adds terms that grow with the residuals. Computed at KAPSAM's own solution, the Gauss–Newton standard uncertainties agree with Suncal's (column 7), so the remaining difference is the definition of the covariance, not the fit.

4. Conformity assessment (JCGM 106)

Project	quantity	KAPSAM	Suncal	relative difference
emue-e6.5-ephedrine-wada-decision-limit	p_c, EQAS 1	0.000122898	0.000122898	4.1e-13
emue-e6.5-ephedrine-wada-decision-limit	p_c, EQAS 2	0.000428958	0.000428958	8.1e-14
emue-e6.5-ephedrine-wada-decision-limit	p_c, EQAS 3	0.00167636	0.00167636	1.4e-14
eurachem-compliance-b1-nickel-steel	p_c, batch	0.841345	0.841345	2.2e-16
eurachem-compliance-b2-upper-limit-t	p_c, batch	0.0655541	0.0655541	8.9e-16
eurachem-compliance-b3-banned-substance-normal	p_c, sample	0.0316454	0.0316454	4.4e-15
jcgm106-7.3-bursting-strength	p_c, container	0.98901	0.98901	0.0e+00
jcgm106-7.3-zener-diode	p_c, diode	0.919243	0.919243	2.2e-16
jcgm106-7.4-motor-oil-viscosity	p_c, oil	0.66263	0.66263	4.4e-16
jcgm106-8.3-speed-limit	p_c, car A	0.00863397	0.00863397	1.1e-15
jcgm106-8.3-speed-limit	p_c, car B	0.000106237	0.000106237	2.8e-13
jcgm106-9.5-ball-bearings	consumer risk	0.001	0.001	1.4e-10
jcgm106-9.5-ball-bearings	producer risk	0.0754939	0.0754939	2.1e-14
jcgm106-9.5-precision-resistors	consumer risk	0.00987829	0.00987829	3.1e-12

Project	quantity	KAPSAM	Suncal	relative difference
jcgm106-9.5-precision-resistors	producer risk	0.0690265	0.0690265	6.1e-13
tur-guard-band-dobbert-tur2-5	p_c, reading 0.70	0.933193	0.933193	2.2e-16
tur-guard-band-dobbert-tur2-5	p_c, reading 0.95	0.598706	0.598706	0.0e+00
tur-guard-band-dobbert-tur2-5	consumer risk	0.00688318	0.00688318	1.1e-12
tur-guard-band-dobbert-tur2-5	producer risk	0.0521624	0.0521624	2.1e-14
tur-guard-band-rp10-tur2-0	p_c, reading 0.70	0.88493	0.88493	2.2e-16
tur-guard-band-rp10-tur2-0	p_c, reading 0.95	0.57926	0.57926	0.0e+00
tur-guard-band-rp10-tur2-0	consumer risk	0.00321313	0.00321313	2.4e-12
tur-guard-band-rp10-tur2-0	producer risk	0.140039	0.140039	7.8e-15
tur-guard-band-rss-tur1-5	p_c, reading 0.70	0.81594	0.81594	2.2e-16
tur-guard-band-rss-tur1-5	p_c, reading 0.95	0.559618	0.559618	2.2e-16
tur-guard-band-rss-tur1-5	consumer risk	0.00567865	0.00567865	9.4e-13
tur-guard-band-rss-tur1-5	producer risk	0.177009	0.177009	2.2e-15
emue-e2.2-vacuum-gauge-correction-applied	p_c, f	0.74533	0.74533	0.0e+00
emue-e2.2-vacuum-gauge-correction-ignored	p_c, f	0.838346	0.838346	2.2e-16
emue-e2.2-vacuum-gauge-correction-in-uncertainty	p_c, f	0.778233	0.778233	0.0e+00
emue-e2.2-vacuum-gauge-sum-model-correlated	p_c, dp	0.745342	0.745342	0.0e+00
emue-e2.2-vacuum-gauge-sum-model-correlation-ignored	p_c, dp	0.669006	0.669006	2.2e-16

max relative difference = 1.4e-10 (32 values)

Monte Carlo, all sections: 79 comparisons, max |z| = 2.4; P(max |z| ≥ 2.4 among 79 independent N(0, 1)) = 0.75

Not compared

Project	reason
gum5-e2-benzo-a-pyrene-ensemble	ensembles or implicit model
nist-tn1900-e11-step-attenuator	Monte Carlo only
nist-tn1900-e13-thermal-expansion-student-95	Monte Carlo only
nist-tn1900-e13-thermal-expansion-student-99	Monte Carlo only
iupac2024-3a-argentometric-titration-ksp	ensembles or implicit model

Project	reason
jcgm102-9.4-resistance-reactance-alternative	ensembles or implicit model
jcgm102-9.4-resistance-reactance	ensembles or implicit model
jcgm102-9.5.2-resistance-thermometer	ensembles or implicit model
jcgm102-9.5.3-ten-temperatures	ensembles or implicit model
emue-e3.3-no2-analyser-dynamic-dilution	covariances between calibration points
iso6143-b2-n2-natural-gas-covariances	covariances between calibration points
eurachem-compliance-b3-banned-substance-lognormal-exact	specific risks: lognormal measurement or posterior with a process prior
eurachem-compliance-b3-banned-substance-lognormal	specific risks: lognormal measurement or posterior with a process prior
gum5-e10-tspm-global-risks-acceptance-0.17	global risks: uncertainty relative to the value
gum5-e10-tspm-global-risks-acceptance-0.22	global risks: uncertainty relative to the value
gum5-e10-tspm-global-risks-shared-risk	global risks: uncertainty relative to the value
gum5-e10-tspm-specific-risks	specific risks: lognormal measurement or posterior with a process prior
gum5-e10-tspm-specific-risks	global risks: uncertainty relative to the value

Monte Carlo propagation through the calibration fit

Generated by `python3 tools/research/benchmarks/calibration_mc.py` (research log D97). For every calibration project of the library the calibration data are drawn 1 000 000 times from normal distributions with their standard uncertainties and covariances, and the ISO 6143 fit is repeated for every draw (Lira, Measurement 44 (2011) 2156). The standard deviations of the parameters are compared with KAPSAM's linear propagation (exact derivatives of the solution, full Hessian; ISO 6143 A.2, A.6, A.7) and with the Gauss–Newton approximation (Suncal's York and ODR covariances); the numerical refit column checks the program's derivatives by central differences of the refit (A.2). The standard error of the Monte Carlo value comes from 20 batches.

Parameters

Project	function	Γ	parameter	u KAPSAM	u numerical refit / KAPSAM − 1	u Gauss–Newton / KAPSAM − 1	s.d. Monte Carlo / KAPSAM − 1	standard error	(mean − b)/u
emue-e3.3-no2-analyser-dynamic-dilution-covariances-ignored	linear	1.27	b0	17.177	-6.9e-06	+0.57%	+0.15%	0.05%	-0.006
		1.27	b1	0.081087	-5.1e-06	+0.75%	+0.13%	0.05%	+0.006
emue-e3.3-no2-analyser-dynamic-dilution	linear	1.27	b0	16.889	-7.2e-06	+0.58%	+0.15%	0.05%	-0.006
		1.27	b1	0.07978	-5.3e-06	+0.77%	+0.12%	0.05%	+0.006

Project	function	Γ	parameter	u KAPSAM	u numerical refit / KAPSAM − 1	u Gauss–Newton / KAPSAM − 1	s.d. Monte Carlo / KAPSAM − 1	standard error	(mean − b)/u
iso6143-b1-no-uv-linear	linear	0.57	b0	0.15716	-1.0e-05	-0.02%	+0.10%	0.07%	-0.014
		0.57	b1	0.48048	-1.3e-05	-0.03%	+0.04%	0.06%	+0.017
iso6143-b2-n2-natural-gas-covariances	linear	1.63	b0	0.0011463	+2.8e-05	+0.01%	+0.02%	0.08%	-0.000
		1.63	b1	2.5497e-08	+1.2e-05	+0.01%	-0.11%	0.08%	+0.001
iso6143-b2-n2-natural-gas-linear	linear	1.63	b0	0.0011458	+2.7e-05	+0.01%	+0.02%	0.08%	-0.000
		1.63	b1	2.4159e-08	+6.6e-06	+0.01%	-0.12%	0.07%	+0.001
iso6143-b2-n2-natural-gas-quadratic	quadratic	0.87	b0	0.0011748	-5.7e-07	+0.00%	+0.04%	0.09%	-0.000
		0.87	b1	5.9011e-08	-1.4e-05	-0.01%	+0.10%	0.07%	-0.001
		0.87	b2	1.8952e-13	+1.9e-05	-0.00%	+0.08%	0.07%	+0.001
iso6143-b3-no-ndir-exponential	exponential	0.35	b0	3.1814	-3.9e-04	+0.00%	+1.43%	0.06%	-0.062
		0.35	b1	3.1703	-3.9e-04	+0.00%	+1.44%	0.07%	+0.062
		0.35	b2	1.2738e-06	-3.8e-04	+0.00%	-0.09%	0.06%	-0.002
iso6143-b3-no-ndir-power	power	1.16	b0	0.018118	+8.9e-05	+0.74%	-0.06%	0.05%	-0.008
		1.16	b1	2.4181e-05	+1.2e-04	+0.69%	+0.13%	0.06%	+0.026
		1.16	b2	0.0050773	+1.2e-04	+0.68%	-0.07%	0.06%	-0.002

Contents of the samples

Project	sample	u(x) KAPSAM	Gauss–Newton / KAPSAM − 1	s.d. Monte Carlo / KAPSAM − 1	standard error
emue-e3.3-no2-analyser-dynamic-dilution-covariances-ignored	validation	4.3503	+0.11%	-0.07%	0.06%
emue-e3.3-no2-analyser-dynamic-dilution	validation	4.5194	+0.10%	-0.07%	0.06%
iso6143-b1-no-uv-linear	S1	0.16377	+0.00%	+0.08%	0.08%
iso6143-b1-no-uv-linear	S2	0.35599	-0.01%	-0.03%	0.08%
iso6143-b1-no-uv-linear	S3	1.1631	-0.01%	+0.15%	0.07%
iso6143-b2-n2-natural-gas-covariances	S1	0.0020926	-0.01%	+0.05%	0.07%

Project	sample	u(x) KAPSAM	Gauss–Newton / KAPSAM − 1	s.d. Monte Carlo / KAPSAM − 1	standard error
iso6143-b2-n2-natural-gas-covariances	S2	0.010406	+0.01%	-0.03%	0.07%
iso6143-b2-n2-natural-gas-linear	S1	0.0020244	-0.01%	+0.05%	0.07%
iso6143-b2-n2-natural-gas-linear	S2	0.0099712	+0.01%	-0.03%	0.07%
iso6143-b2-n2-natural-gas-quadratic	S1	0.0032911	-0.02%	+0.08%	0.07%
iso6143-b2-n2-natural-gas-quadratic	S2	0.011761	+0.02%	-0.03%	0.06%
iso6143-b3-no-ndir-exponential	S1	0.014246	+0.00%	-0.04%	0.06%
iso6143-b3-no-ndir-power	S1	0.014171	+0.07%	-0.03%	0.06%

Draws for which the fit did not converge: emue-e3.3-no2-analyser-dynamic-dilution-covariances-ignored 0, emue-e3.3-no2-analyser-dynamic-dilution 0, iso6143-b1-no-uv-linear 0, iso6143-b2-n2-natural-gas-covariances 0, iso6143-b2-n2-natural-gas-linear 0, iso6143-b2-n2-natural-gas-quadratic 0, iso6143-b3-no-ndir-exponential 0, iso6143-b3-no-ndir-power 0.